Abdelouahed Ben Mhamed

Ph.D. Student in Operations Research & Machine Learning
AI Movement, International Artificial Intelligence Center of Morocco,
Mohammed VI Polytechnic University (UM6P)
Rabat Campus, Rocade Rabat-Salé, 11103 Rabat, Morocco
+212-614-592-144 · abdelouahed.benmhamed@um6p.ma
ben-mhamed.github.io · LinkedIn · Google Scholar · GitHub

Education

Ph.D. in Operations Research and Machine Learning, UM6P, Rabat, Morocco 2023 – Present
Thesis: *Machine Learning for Solving Combinatorial Optimization: Application to Drone Routing Problems*
Thesis supervisors: Dr. Assia Kamal-Idrissi and Prof. Amal El Fallah-Seghrouchni

Engineering Degree (Ing.) in Operational Research, INSEA, Rabat, Morocco 2009 – 2012
Bac+5, equivalent to an M.S.

D.E.U.G. in Mathematics and Applications, Meknès, Morocco 2007 – 2009
Faculty of Sciences, Moulay Ismail University
Two-year undergraduate degree (Diplôme d'Études Universitaires Générales)

Research Interests

- Machine learning for combinatorial optimization
- Drone and vehicle routing problems (TSP-D, truck–drone logistics)
- Generative models for discrete structures (GFlowNets)
- Decision diagrams and learning-augmented exact methods
- Reinforcement learning for solver decisions (branching, cutting planes)

Publications in Machine Learning & Optimization

Peer-Reviewed

Ben Mhamed, A., Kamal-Idrissi, A., & El Fallah-Seghrouchni, A. (2026). Learning branching policies for MILPs with proximal policy optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*, *40*, 24379–24387.

Under Review

Ben Mhamed, A., Ajaoud, H., Kamal-Idrissi, A., & El Fallah-Seghrouchni, A. (2026). Learning to compile relaxed decision diagrams by optimizing the bound. Under review at *ICLR 2027*.

Ben Mhamed, A., Kamal-Idrissi, A., & El Fallah-Seghrouchni, A. (2026). What matters in learned pipelines for the traveling salesman problem with drone: search objective, start diversity, and learned priors. Under review at *Transportation Research Part E* (special issue: From Learning to Optimization in Intelligent Logistics Systems).

Ajaoud, H., **Ben Mhamed, A.**, Kamal-Idrissi, A., & El Fallah-Seghrouchni, A. (2026). Bellman-state decision diagrams for the multi-destination integrated last-mile transportation problem. Under review at *AAAI 2027*.

Presentations

Ben Mhamed, A., Kamal-Idrissi, A., & El Fallah-Seghrouchni, A. (2026). *Learning branching policies for MILPs with proximal policy optimization* [Poster]. *AAAI Conference on Artificial Intelligence*, Singapore.

Research Experience

Doctoral Researcher, AI Movement, UM6P, Rabat, Morocco Sep 2023 – Present
Supervisors: Dr. Assia Kamal-Idrissi and Prof. Amal El Fallah-Seghrouchni

- Develop learning-augmented methods for truck–drone routing: operation-level GFlowNets, decision-diagram-based bounds, and learned branching and cutting policies inside exact MILP solvers.
- Design evaluation protocols with matched computational budgets against strong classical baselines (LKH, local search, partition DP).
- First-author publication at AAAI 2026; three manuscripts under review (ICLR 2027, Transportation Research Part E, AAAI 2027).

Statistical Modeller, INRH, Casablanca, Morocco Jan 2013 – Jan 2019
National Institute of Fisheries Research (government research institute)

- Built stock-assessment models used to set national fishing quotas, including the *DEPLr* R package adopted for Morocco's octopus quota decisions.
- Developed a modified dynamic biomass model for short-lived species (*Biodysls* R package) and climate-driven landing models for the artisanal sardine fishery.
- Contributed to internationally funded programmes: bluefin tuna stock evaluation with the state-space SAM model (European Commission) and power/cost–benefit analysis for the ICCAT-GBYP aerial survey.
- Co-authored seven peer-reviewed papers and six technical reports in marine and fisheries science.

R Consultant, IRD, Casablanca, Morocco Sep 2012 – Dec 2012
French National Research Institute for Sustainable Development

- Built an optimization model allocating Moroccan sardine landings across economic transformation sectors.

Industry Experience

Data Scientist (part-time, remote), CIEK Solutions, United States 2016 – 2025
Marketing analytics, held in parallel with the roles below

- Developed marketing mix models quantifying sales and profit response to media investment for retail and food-service clients, including The Container Store, Eyemart and McDonald's.
- Formulated and solved media-budget optimization problems, allocating spend across channels to maximize sales or profit under budget constraints.

Founder, Manager & Data Scientist, INTELLIGENTICA, Casablanca, Morocco Nov 2020 – Feb 2024
Led machine-learning projects for public- and private-sector clients

- *Syngenta Morocco*: led a machine-learning project for anomaly detection in the client's operational data.
- *FISHEYE*: web platform predicting small-pelagic catches by species, zone and time step, and forecasting their spatial distribution from environmental, biological and oceanographic indicators.
- *OCEAN VIEW*: centralised INRH's data into a national fisheries information system and migrated hosting to a cloud architecture.

– *SMARK*: web platform for marketing mix modelling and optimization of media-budget allocation across channels.

Data Scientist, CAPFI, Casablanca, Morocco Oct 2019 – Sep 2020

– Designed, trained and validated predictive models for client business problems, from problem framing and feature engineering to performance evaluation.
– Built data-preparation pipelines and deployed models into production workflows, with monitoring of their performance over time.
– Presented analytical results and recommendations to client stakeholders and advised junior team members on modelling practice.

Data Scientist, Agil-IT, Paris, France Jan 2019 – Sep 2019

– Delivered data-science consulting assignments for client organisations, translating business needs into statistical and machine-learning solutions.
– Performed exploratory analysis, statistical modelling and forecasting on large structured datasets in Python and R.
– Produced dashboards and technical reports communicating model results to non-technical audiences.

Teaching Experience

Instructor, MLOps for Mining: Scaling AI from Pilot to Production 2026
Professional training for the mining industry, UM6P

– Taught how to take machine-learning systems from pilot to production in an industrial setting.

Instructor, Practical Sessions in Optimization with Gurobi 2026
Executive Master in Artificial Intelligence, UM6P

– Hands-on sessions on modelling and solving optimization problems with the Gurobi solver.

Instructor, Junior Master in AI (ages 8–14) 2026
National "AI for Kids" Program, Ministry of Digital Transition and Administrative Reform, Morocco

– Taught artificial-intelligence fundamentals to children aged 8–14 within the national initiative.

Instructor, Introduction to R Programming for Fisheries Scientists, INRH 2017

– Designed and delivered an R programming course tailored to fisheries research workflows.

Publications in Marine & Fisheries Science

Peer-Reviewed

Dridi, A., Srairi, A., Boumaaz, A., Bensbai, J., **Ben Mhamed, A.**, & Belghyti, D. (2022). Study of reproduction of the common octopus *Octopus vulgaris* in the South Atlantic area of Morocco (1998–2010). *AACL Bioflux*, *15*(5), 2374–2387.

Chaira, K., Rhinane, H., Ennaffah, B., Maimouni, S., Sagou, R., Loulad, S., **Ben Mhamed, A.**, Agouzouk, A., BenBrahim, S., Masseret, E., & Laabir, M. (2021). The distribution of dinoflagellate cyst assemblages in recent sediments of the Oualidia Lagoon, Morocco, with a focus on toxic species. *African Journal of Marine Science*, *43*(3), 279–292. doi.org/10.2989/1814232X.2021.1945684

Somoue, L., Demarcq, H., Makaoui, A., Hilmi, K., Ettahiri, O., **Ben Mhamed, A.**, Agouzouk, A., Baibai, T.,

Larissi, J., Charib, S., Kalmouni, A., & Laabir, M. (2020). Influence of ocean–lagoon exchanges on spatio-temporal variations of phytoplankton assemblage in an Atlantic lagoon ecosystem (Oualidia, Morocco). *Regional Studies in Marine Science*. doi.org/10.1016/j.rsma.2020.101512

Ben Gharbia, H., Laabir, M., **Ben Mhamed, A.**, Maïté Gueroun, S. K., Daly Yahia, M. N., Nouri, H., M'Rabet, C., Shili, A., & Kéfi-Daly Yahia, O. (2019). Occurrence of epibenthic dinoflagellates in relation to biotic substrates and to environmental factors in Southern Mediterranean (Bizerte Bay and Lagoon, Tunisia). *Harmful Algae*, *90*.

Oumarous, M., Lakhnigue, A., **Ben Mhamed, A.**, Bensbai, J., Charouki, N., Malouli, M., & Bekkali, M. (2017). Artisanal fishery in Moroccan North Atlantic: case of small pelagic. *Ecology, Environment and Conservation*, *24*(2), 36–47.

Oumarous, M., Lakhnigue, A., **Ben Mhamed, A.**, Houssa, R., Charouki, N., Malouli, M., & Bekkali, M. (2016). Artisanal fishery of small pelagic: diagnosis and prospects, case of Moroccan Atlantic centre. *International Journal of Advanced Research*, *4*(7), 1372–1386.

Wahbi, F., Le Loc'h, F., Berraho, A., Benazzouz, A., **Ben Mhamed, A.**, & Errhif, A. (2015). Composition et variations spatio-temporelles du régime alimentaire de *Trachurus trachurus* de la côte atlantique marocaine. *Cybium*, *39*(2), 131–142.

Technical Reports & Working Documents (non-peer-reviewed)

Kell, L. T., **Ben Mhamed, A.**, Rouyer, T., & Kimoto, A. (2018). An evaluation of bias and prediction skill for the East Atlantic bluefin stock assessment. *Collective Volume of Scientific Papers ICCAT*, *74*(6), 3052–3081. (SCRS/2017/124)

Cañadas, A., & **Ben Mhamed, A.** (2016). Power analysis and cost–benefit analysis for the ICCAT GBYP aerial survey on bluefin tuna spawning aggregations. Final report, ICCAT Atlantic-Wide Research Programme on Bluefin Tuna (GBYP), Phase 5, Tender No. ICCAT-GBYP 08/2015.

Baibbat, S. A., Abid, N., Malouli, M. I., **Ben Mhamed, A.**, & Ikkiss, A. (2017). Updated standardized catch rate of swordfish (*Xiphias gladius*) from the Moroccan longline fishery operating south of the Moroccan Atlantic waters. *Collective Volume of Scientific Papers ICCAT*, *74*(3), 1000–1008. (SCRS/2017/063)

Abid, N., Baibbat, S. A., & **Ben Mhamed, A.** (2015). Updated standardized catch rate of the bigeye tuna (*Thunnus obesus*) from the Moroccan longline fishery operating in the Atlantic. ICCAT SCRS document SCRS/2015/075.

Abid, N., Malouli, M., Bensbai, J., **Ben Mhamed, A.**, Benchoucha, S., El Arraf, S., & El Fanichi, C. (2015). Standardized CPUE of bluefin tuna (*Thunnus thynnus*) caught by Moroccan traps for the period 1986–2014. *Collective Volume of Scientific Papers ICCAT*, *71*(4), 1632–1638. (SCRS/2014/168)

Talbaoui, E. M., Bensbai, J., **Ben Mhamed, A.**, ..., & Camiñas, J. A. (2014). Report of the 3rd meeting of the CopeMed II working group between Spain and Morocco on blackspot seabream (*Pagellus bogaraveo*) of the Strait of Gibraltar area. FAO CopeMed II Technical Documents No. 37 (GCP/INT/028/SPA – GCP/INT/006/EC).

Service & Outreach

Member, Moroccan Scientific Delegation 2013 – 2019
International Commission for the Conservation of Atlantic Tunas (ICCAT), FAO, and General Fisheries Commission for the Mediterranean (GFCM-FAO)

- Represented INRH at ICCAT's Standing Committee on Research and Statistics (SCRS) and its methods and stock-assessment working groups for bluefin tuna, swordfish and bigeye tuna, contributing quantitative

analyses to the scientific advice underpinning international fishery management.
- Took part in the FAO CopeMed II Spain–Morocco working group on blackspot seabream in the Strait of Gibraltar.
- Co-authored six technical reports and SCRS working documents on stock-assessment diagnostics, abundance-index standardization and survey design.

Mentor, AI Hackathon 2026
Ministry of Digital Transition and Administrative Reform, Morocco
- Mentored participating teams on problem framing, modelling and machine-learning methodology.

Software & Research Tools

DEPLr (R package) — octopus stock assessment; used to establish fishing quotas in Morocco.

Biodysls (R package) — modified dynamic biomass model tailored to short-lived species.

FISHEYE (web platform) — machine-learning prediction of small-pelagic catches and their spatial distribution.

SMARK (web platform) — marketing mix modelling and optimization of media-budget allocation across channels.

OCEAN VIEW (data platform) — national fisheries information system and cloud data pipelines for INRH.

Technical Skills

Programming: Python, R, C/C++, Scala, SQL, JAGS, WinBUGS

ML & Optimization: PyTorch, reinforcement learning, GFlowNets, Gurobi, MILP modelling, decision diagrams

Data & Cloud: PostgreSQL, MySQL, SQLite, Google Cloud Platform

Web: Flask, ReactJS, Shiny, HTML5/CSS3, JavaScript

Languages: Arabic (native), French (proficient), English (proficient)

Referees

Dr. Assia Kamal-Idrissi
Primary Ph.D. Supervisor
UM6P
Operations Research
assia.kamal-idrissi@um6p.ma

Prof. Amal El Fallah-Seghrouchni
Ph.D. Co-Supervisor
UM6P
Artificial Intelligence
amal.elfallah-seghrouchni@um6p.ma

Prof. Laurence T. Kell
Imperial College London
ICCAT SCRS co-author
Fisheries Stock Assessment
l.kell@imperial.ac.uk

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